© CFA Institute. For candidate use only. Not for distribution.

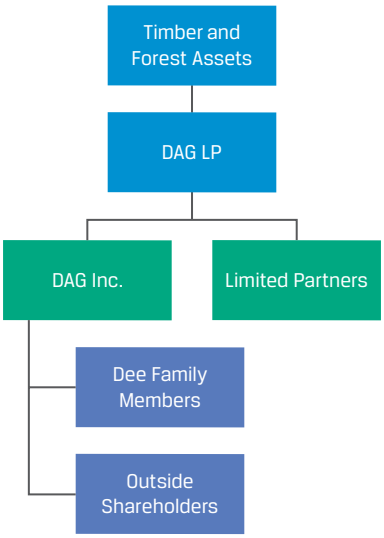
PRACTICE PROBLEMS

The following information relates to questions 1-5

Dee's Arbor Group Inc. (DAG Inc.) is a large international investor in timber and forest assets located on the North and South American continents. DAG Inc. is the general partner of DAG LP, a limited partnership that is the 100% owner of the timber and forest assets. DAG LP is controlled by DAG Inc. and its limited partners; the limited partners own a 20% stake in the partnership, while DAG Inc. holds the remaining majority stake.

DAG Inc.'s shares are listed on stock exchanges in the United States and Canada. DAG Inc. is organized as a special corporate form available in its jurisdiction in which it does not pay corporate income taxes so long as it distributes all of its net income as dividends to its shareholders and complies with other conditions. In the current and past fiscal years, DAG Inc. has complied with all of these conditions.

DAG Inc.'s shares are owned by various members of the Dee family, who hold several key senior management positions at DAG Inc. and DAG LP, and collectively they own 30% of the shares of DAG Inc. The remainder of DAG Inc.'s shares are owned by a variety of individual and institutional investors, none of whom own more than 5%. The following diagram shows the organizational structure of DAG Inc. and DAG LP.



1. What percentage of the timber and forest assets are effectively owned by the Dee family members in this scenario?

   A. 24%

   B. 30%

   C. 80%

2. Which of the following best describes the taxation of DAG Inc. and DAG LP?

   A. DAG LP pays tax based on its pre-tax income.

© CFA Institute. For candidate use only. Not for distribution.

B. Shareholders of DAG Inc. pay tax based on dividend income.

C. DAG Inc. pays tax based on its pre-tax income.

3. Which of the following best describes DAG LP and DAG Inc.?

A. Neither DAG LP nor DAG Inc. is managed by a majority owner.

B. DAG Inc. has limits to its ability to raise new capital because of the dividend requirement on its corporate form.

C. DAG Inc. faces unlimited liability as the general partner, while DAG Inc.'s status as a corporation implies that its shareholders face limited liability.

4. What is a significant difference between the limited partners of DAG LP and the outside shareholders of DAG Inc.?

A. Managerial responsibilities

B. Taxation of income from the partnership

C. The ability to vote and replace members of the DAG Inc. board of directors

5. Which of the following rationales would be most consistent with the Dee family's choice to create this complex organizational structure instead of simply organizing as a limited partnership?

A. It provides management control without the need for majority ownership by the family, while maintaining limited liability.

B. The complex structure eliminates the potential for double taxation.

C. The complex structure avoids potential problems associated with outside investors exercising voting rights.

© CFA Institute. For candidate use only. Not for distribution.

SOLUTIONS

1. A is correct. The Dee family effectively owns 24% of the timber and forest assets. They own 30% of the shares of DAG (the publicly traded corporation), and they own 80% of the limited partnership (which owns the assets): 24% = 30% × 80%.

2. B is correct. DAG Inc. is organized as a special corporate form available in its jurisdiction in which it does not pay corporate income taxes. DAG LP, as a limited partnership, is a pass-through entity. DAG Inc. shareholders pay tax on dividend income.

   A is incorrect. DAG LP is a limited partnership. Partnership income is passed through to each partner, and the partners pay tax at the personal level.

   C is incorrect. DAG Inc. is organized as a special corporate form available in its jurisdiction in which it does not pay corporate income taxes.

3. C is correct. DAG Inc. is the general partner of DAG LP and thus has unlimited liability in the partnership. However, as a corporation, DAG Inc.'s shareholders have limited liability for its losses and obligations.

   A is incorrect. DAG LP is 80% owned by DAG Inc., the general partner that manages the partnership. DAG Inc. is 30% owned by the Dee family, who holds several key managerial positions.

   B is incorrect. As a public company, DAG Inc. can raise new equity by issuing shares on an exchange. The dividend requirement is not a binding limit on its ability to raise capital but, rather, might be an attractive feature to prospective investors.

4. C is correct. The limited partners of DAG LP are not shareholders of DAG Inc., so they do not have voting rights in the corporation. Additionally, because the limited partners own only 20% of the partnership, they also have little ability to remove DAG Inc. as the general partner. In contrast, outside shareholders own 70% of DAG Inc. and have voting rights. They could use their collective ownership to effect change in the management of DAG Inc.

   A is incorrect. Neither the DAG LP limited partners nor the outside shareholders of DAG Inc. have managerial responsibilities. The general partner has managerial responsibilities of the partnership, and the board of DAG Inc. has managerial responsibilities of the corporation.

   B is incorrect. Owing to the special corporate form of DAG Inc., both the corporation *and* the limited partnership are pass-through entities. Therefore, neither the partnership nor the corporation pays entity-level income taxes, but both the limited partners and shareholders are responsible for personal income taxes.

5. A is correct. The Dee family can effectively control management of the timber and forest assets with only a 24% effective ownership stake in the partnership. If the family had opted for a limited partnership as the organizational form, they likely would need a much higher ownership stake to assert management control and would have unlimited personal liability.

   B is incorrect. The double taxation problem is not an issue in the more complex structure, nor is it a problem in the partnership structure, because both are pass-throughs.

   C is incorrect. The complex structure potentially creates problems because of the existence of outside shareholders constituting a majority of votes.

© CFA Institute. For candidate use only. Not for distribution.

PRACTICE PROBLEMS

The following information relates to questions 1-3

Consider a firm with assets of 200 fully financed by equity and an identical firm financed with 80% debt and 20% equity. Assume both firms have revenue over a period of 200 and non-debt operating expenses of 140.

1. If we assume an interest rate of 20% for the period, calculate net income for each firm and compare their returns on equity for the period, ignoring income taxes.

2. Calculate the return on equity for the period if the firms experience a 20% increase and decrease in revenue (from Question 1), assuming operating expenses remain unchanged.

3. Consider the same all-equity-financed firm as in Question 1 and its choices for financing a new investment in LT assets of 40. The pertinent details in the firm's initial balance sheet are shown below. Revenue before the investment is 200, operating expenses are 140 and are expected to remain unchanged, interest on new debt financing is 20%, and the return on the new investment is 30%.

   Calculate the ROE for the firm if it finances the investment with debt, the issuance of new shares, or cash on hand and compare the results. Discuss the results of the comparison.

Initial Balance Sheet			
Cash	60		
Other assets	40		
LT assets	100	Equity	200

Share Issuance		Debt Issuance		Cash on Hand	
Cash 60		Cash 60		Cash 20	
Other 40		Other 40	Debt 40	Other 40	
LT assets 140	Equity 240	LT assets 140	Equity 200	LT assets 140	Equity 200

Revenue	212	212	212
Less: Operating Expenses	140	140	140
Less: Interest		8	
Net Income	72	64	72
Total Equity	240	200	200
ROE	30%	32%	36%

4. A company has developed a long-term relationship with its major supplier. The supplier has developed complex systems that integrate with those of the compa-

© CFA Institute. For candidate use only. Not for distribution.

ny and has an agreement to receive payment within 30 days for goods delivered. Discuss the supplier's stakeholder relationship with the firm. Why would high financial leverage be inconsistent with the supplier's interests?

5. Explain why a company's management might not act in the best interests of shareholders.

© CFA Institute. For candidate use only. Not for distribution.

SOLUTIONS

1. Solve for net income by subtracting expenses from revenue, and divide net income by total equity to solve for the one-period return on equity:

Equity-Financed		Debt- and Equity-Financed	
Revenue	200	Revenue	200
Less: Operating Expenses	–140	Less: Operating Expenses	–140
		Less: Interest Expense*	–32
Net Income	60	Net Income	28
Total Equity	200	Total Equity	40
Return on Equity	**30%**	**Return on Equity**	**70%**

*Interest expense with 80% debt financing = 160 × 0.20 = 32.

2.

20% *Increase* in Revenue

Equity-Financed		Debt- and Equity-Financed	
Revenue	240	Revenue	240
Less: Operating Expenses	–140	Less: Operating Expenses	–140
		Less: Interest Expense	–32
Net Income	100	Net Income	68
Total Equity	200	Total Equity	40
Return on Equity	**50%**	**Return on Equity**	**170%**

20% *Decrease* in Revenue

Equity-Financed		Debt- and Equity-Financed	
Revenue	160	Revenue	160
Less: Operating Expenses	–140	Less: Operating Expenses	–140
		Less: Interest Expense	–32
Net Income	20	Net Income	**(12)**
Total Equity	200	Total Equity	40
Return on Equity	**10%**	**Return on Equity**	**(30%)**

Like the example in the lesson, the use of debt increases the variance in return on equity.

3. Financing the investment with new shares produces the lowest ROE among the three financing options. However, since the investment produced a return equal

© CFA Institute. For candidate use only. Not for distribution.

to the beginning ROE, there was no change in ROE from the initial case. Financing with debt produced a higher ROE, because the interest rate on debt was lower than the return on the new investment and no new shares were issued, avoiding dilution. The highest ROE was produced by using excess cash on hand, which avoids both dilution and the interest cost of debt.

4. The supplier has a clear vested interest in the success and survival of the company, so it is considered a stakeholder. The supplier extends short-term credit to the company and has a long-term interest in the success of the company because it has invested in integrated complex systems. High financial leverage would not be in the best interest of the supplier, because it presents both short-term risk that the company may be unable to make payments for goods delivered and long-term risk for the investment made to integrate the supplier's complex systems with those of the company.

5. Managers are in a principal-agent relationship with a company's shareholders. Managers, overseen by the board of directors, serve as agents who should act in the best interests of the company's shareholders. However, in some cases, managers may put their own interests ahead of shareholders' interests. Examples include insufficient effort, excessive perquisite consumption (e.g., corporate jets, elaborate offices), and failure to take appropriate risks or make investments in an effort to safeguard their jobs. Oversight by a majority independent board and compensation to align managers' interests with shareholders', such as performance-based and equity-based compensation schemes, are mechanisms to mitigate these conflicts of interest.

© CFA Institute. For candidate use only. Not for distribution.

PRACTICE PROBLEMS

The following information relates to questions 1-5

Kobe Steel Ltd. is a major Japanese steel manufacturer formed in 1905 that supplies global manufacturers of cars, planes, and trains. The company issued a report in March 2018 apologizing for falsifying data on the strength and durability of its aluminum, copper, steel products, and iron ore powder. Following the scandal, its stock sank to a five-year low and its cost of debt jumped to record levels.

An Independent Investigation Committee (IIC) investigated the misconduct and authored the report, which concluded that the misconduct resulted from (1) a management style that overemphasized profitability amid inadequate corporate governance, (2) the imbalanced operation of plants that resulted in reduced awareness of quality compliance among employees, and (3) insufficient quality control procedures that allowed the misconduct to take place. The report also emphasized that the company had "a culture that prioritized winning purchase orders and meeting delivery deadlines over ensuring quality."

The company committed to implementing various measures to prevent a recurrence of the misconduct. Regarding governance, measures included the following:

- Appointing an independent chairman of the board of directors
- Having at least one-third of the board of directors be independent
- Creating a nominating and compensation committee, consisting of a majority of independent directors, to serve as a voluntary advisory body to the board of directors
- Creating an independent quality supervision committee consisting of external experts
- Creating an audit and supervisory committee consisting of five members: two internal directors and three independent directors with backgrounds in legal, financial, and industrial fields

1. Identify three key stakeholder relationships for Kobe Steel and discuss their role in the misconduct or how they were affected.

2. Discuss the stakeholder incentives and the conflicts that arose between Kobe's management and the company's customers.

3. Which of the following is considered a best practice that would strengthen the measures discussed to prevent a recurrence of the misconduct?

   A. A shareholder rights plan

   B. A stock-based compensation plan

   C. An audit committee composed solely of independent board members

4. Discuss the financial risk implications of the post-scandal stock and bond prices

© CFA Institute. For candidate use only. Not for distribution.

with regard to investor confidence.

5. Kobe Steel's governance failures *most likely* resulted in reduced:

A. cost of debt.

B. agency costs.

C. growth opportunities.

© CFA Institute. For candidate use only. Not for distribution.

SOLUTIONS

1.

Stakeholder 1	Stakeholder 2	Relationship Type	Role or Impact
Kobe Corporation	Customers (manufacturers of cars, planes, and trains)	Contractual	The company delivered an inferior-quality product that could have resulted in harm to the public.
Board of Directors	Kobe Corporation	Agent	The board failed to monitor the activities of the corporation, resulting in reputational damage, civil complaints, and legal action.
Managers	Employees	Contractual	The culture prioritized winning purchase orders and meeting delivery deadlines over ensuring quality.

2. The company supplied potentially faulty steel and other material inputs to manufacturers of cars, planes, and trains. The customers were incentivized by the advertised quality of the products, while management was incentivized by winning purchase orders and meeting delivery deadlines. A certain quality standard should have aligned interests, but instead the potential for profits created a conflict between the directors and customers.

3. C is correct. The company formed an audit and supervisory committee that included a majority of independent directors, but best practices would recommend an audit committee composed solely of independent members, with at least one director with accounting or related financial management experience.

   A is incorrect. A shareholder rights plan, also known as a poison pill, is used to defend against unwanted takeovers that could potentially undervalue the company, but it does not defend the company or shareholders against management or employee misconduct.

   B is incorrect. A stock-based compensation plan is used to align the interests of management with those of shareholders, though it might further incentivize risk-taking and misconduct, which is the issue in this case.

4. The stock price fell and the cost of debt increased, indicating that investor and creditor confidence in the company declined. Investors will have to assess how much the company's financial results benefited from the past misconduct and estimate the impact from (1) the company no longer falsifying data and (2) the loss of customer confidence even if the company fixes the issues. Strengthening governance to include best practices is a first step toward restoring investor confidence, but the impact on future sales and profitability is key to the stock price and cost of debt going forward.

© CFA Institute. For candidate use only. Not for distribution.

5. C is correct. The misconduct likely caused reputational damage to Kobe among current and prospective customers, which reduces growth opportunities.

   A is incorrect, because bond yields jumped to record levels soon after the scandal was announced, suggesting a higher perceived default risk. Higher default risks are associated with a higher cost of debt, as creditors typically require a higher return when their funds are less secured and their rights less protected.

   B is incorrect, because Kobe is incurring higher agency costs in the wake of the governance failure by hiring an independent investigative committee, independent directors, and external quality assurance experts.

© CFA Institute. For candidate use only. Not for distribution.

PRACTICE PROBLEMS

1. An issuer changing its credit terms for customers from 2/10, net 30 to 2/10, net 40 will most likely experience:

   A. a pull on its liquidity.

   B. a drag on its liquidity.

   C. no change in its liquidity.

2. Which of the following will most likely *decrease* an issuer's cash conversion cycle? An increase in its days:

   A. sales outstanding.

   B. of inventory on hand.

   C. payable outstanding.

3. An analyst gathers balance sheet information for the most recent fiscal year for three issuers.

	Issuer A	Issuer B	Issuer C
Cash	100	120	50
Marketable securities	20	10	20
Accounts receivable	300	300	200
Inventory	500	600	300
Prepaid expenses	50	0	10
Accounts payable	400	500	300
Accrued expenses	40	20	0

   Which issuer is most liquid based on the quick ratio?

   A. Issuer A

   B. Issuer B

   C. Issuer C

4. An analyst is evaluating an issuer's liquidity and calculates a negative cash conversion cycle for the issuer in the most recent fiscal year. This result is:

   A. not feasible.

   B. possible, because the issuer has sufficient cash and marketable securities on hand to support short-term needs.

   C. possible, because the investment in inventory is returned quickly and the issuer takes advantage of vendor financing.

5. An issuer eliminated the prompt-payment discount it had offered to customers. This action most likely will:

   A. increase the issuer's liquidity.

© CFA Institute. For candidate use only. Not for distribution.

B. decrease the issuer's liquidity.

C. not affect the issuer's liquidity.

© CFA Institute. For candidate use only. Not for distribution.

SOLUTIONS

1. B is correct. By extending the net period, the issuer will likely see its accounts receivable increase, lengthening its cash conversion cycle and producing a drag on its liquidity.

   A is incorrect, because the change will slow payments from customers (inflows) rather than pull on liquidity in terms of outflows.

   C is incorrect, because extending the net period will slow payments from customers and produce a drag on liquidity.

2. C is correct. Increasing days payable outstanding would reduce the cash conversion cycle, because payments to suppliers are delayed. Days payable outstanding is subtracted from the sum of days sales outstanding and days of inventory on hand to compute the cash conversion cycle.

   A is incorrect, because an increase in days sales outstanding will increase the issuer's cash conversion cycle.

   B is incorrect, because an increase in days of inventory on hand will increase the issuer's cash conversion cycle.

3. A is correct. The quick ratio is defined as:

   $$\text{Quick ratio} = \frac{\text{Cash + Short-term marketable instruments + Receivables}}{\text{Current liabilities}}$$

   The quick ratio for each issuer is:

   $$\text{Issuer A quick ratio} = \frac{100 + 20 + 300}{400 + 40} = \frac{420}{440} = 0.9545$$

   $$\text{Issuer B quick ratio} = \frac{120 + 10 + 300}{500 + 20} = \frac{430}{520} = 0.8269$$

   $$\text{Issuer C quick ratio} = \frac{50 + 20 + 200}{300 + 0} = \frac{270}{300} = 0.9000$$

4. C is correct. If days of inventory on hand is low, the accounts receivable collection period is short, and the issuer takes advantage of its vendors' financing, a negative cash conversion cycle is a possible result.

   A is incorrect, because issuers may have a negative cash conversion cycle if days payable outstanding is larger than the sum of days of inventory on hand and days sales outstanding.

   B is incorrect, because the amount of cash and marketable securities on hand does not affect the company's cash conversion cycle.

5. B is correct, because the elimination of the discount will likely result in customers paying later as there is no incentive to pay early without the discount. With customers paying later, this action becomes a drag on liquidity.

   A and C are incorrect, because customers likely paying later will reduce the issuer's liquidity.

© CFA Institute. For candidate use only. Not for distribution.

PRACTICE PROBLEMS

The following information relates to questions 1-5

Larissa Soroka, an analyst at ABC company, has been asked to prepare cash flow forecasts for two mutually exclusive investment projects related to new products. ABC's management asks Soroka to include the cost of market research that was recently completed as well as the potential loss of revenue from existing products that could occur if either project is undertaken. Soroka presents the forecast in Exhibit 1 and is asked to calculate IRR and NPV for both projects. Investors in ABC have a required rate of return of 8%.

Exhibit 1

Cash Flow Forecast for Projects A and B

Time	0	1	2	3	4
Project A	(18.5)	4.5	6.0	6.0	5.5
Project B	(33.5)	(2.5)	(1.0)	24.0	25.5

ABC's management asks Soroka to consider in her forecast the impact of a six-month delay in all future cash flows and estimate the impact of that event on IRR and NPV.

After reviewing the forecast, ABC's management asks Soroka to estimate ABC's ROIC for 20X2. ABC earned operating profit of $8,830, had an effective tax rate of 22%, and reported the balance sheet in Exhibit 2.

Exhibit 2

ABC Balance Sheet
(end-of-period values)

Assets:	20X1	20X2
Cash	1,600	1,720
Short-term assets	30,450	29,910
Long-term assets	60,250	62,060
Total assets	92,300	93,690

Liabilities and equity:	20X1	20X2
Accounts payable	12,930	11,620
Short-term debt	8,030	8,390
Long-term debt	25,040	25,910
Share capital	39,800	40,990

© CFA Institute. For candidate use only. Not for distribution.

Liabilities and equity:	20X1	20X2
Retained earnings	6,500	6,780
Total liabilities and equity	92,300	93,690

ABC's management has an ROIC objective of at least 9.2% and asks Soroka for her recommendation about the projects.

1. When preparing the cash-flow forecasts for both projects, Soroka should include:
   - **A.** only the cost of the market research.
   - **B.** only the loss of revenue from existing products.
   - **C.** both the market research cost and the loss of revenue from existing products.

2. With regard to Project A and Project B, which of the following is true?
   - **A.** Both projects should be invested in according to the IRR decision rule.
   - **B.** Both projects should be invested in according to the NPV decision rule.
   - **C.** Only Project B should be invested in according to the IRR decision rule.

3. The *most likely* impact from the cash-flow timing considered by Soroka is that:
   - **A.** both IRR and NPV would decrease.
   - **B.** both IRR and NPV stay unchanged.
   - **C.** only IRR would decrease but NPV would increase.

4. Based on ABC's balance sheet presented in Exhibit 2, Soroka should calculate:
   - **A.** average invested capital of $72,510 and ROIC of 9.50%.
   - **B.** average invested capital of $80,720 and ROIC of 10.94%.
   - **C.** average invested capital of $92,995 and ROIC of 9.50%.

5. Considering ABC management's ROIC-based investment criterion and the IRR and NPV of both projects, Soroka should recommend to:
   - **A.** reject both projects A and B.
   - **B.** invest only in Project B because it has a higher NPV.
   - **C.** invest only in Project A because has positive cash flow during all four years.

© CFA Institute. For candidate use only. Not for distribution.

SOLUTIONS

1. B is correct. Capital allocation analysis should include only incremental cash flows associated with a new investment. The loss of revenue from existing products is an incremental negative effect that should be included in the analysis. The market research costs in this case are a sunk cost because the research has already been completed and therefore does not affect cash-flow estimates, no matter whether ABC undertakes either of these projects or neither of them.

2. C is correct. The IRR decision rule is to invest if the IRR exceeds the required rate of return or hurdle rate. Project B's IRR is 8.97%, which exceeds the required rate of return of 8%; however, Project A's IRR is 7.04%, which is below the required rate of return of 8%.

   The NPV of both projects can be calculated as follows:

   $$\text{NPV}_A = -18.5 + \frac{4.5}{(1+0.08)^1} + \frac{6}{(1+0.08)^2} \frac{6}{(1+0.08)^3} \frac{5.5}{(1+0.08)^4} = -0.38.$$

   $$\text{NPV}_B = -33.5 + \frac{-2.5}{(1+0.08)^1} + \frac{-2.5}{(1+0.08)^2} \frac{24}{(1+0.08)^3} \frac{25.5}{(1+0.08)^4} = 1.12.$$

   Using the same equation, we can solve for the IRRs by assuming that NPV = 0.

   $$0 = -18.5 + \frac{4.5}{(1+\text{IRR}_\text{A})^1} + \frac{6}{(1+\text{IRR}_\text{A})^2} + \frac{6}{(1+\text{IRR}_\text{A})^3} + \frac{5.5}{(1+\text{IRR}_\text{A})^4}.$$

   $$\text{IRR}_A = 7.04\%.$$

   $$0 = -33.5 + \frac{-2.5}{(1+\text{IRR}_\text{B})^1} + \frac{-2.5}{(1+\text{IRR}_\text{B})^2} + \frac{24}{(1+\text{IRR}_\text{B})^3} + \frac{25.5}{(1+\text{IRR}_\text{B})^4}.$$

   $$\text{IRR}_B = 8.97\%.$$

   A is incorrect because Project B's IRR is 8.97%, which exceeds the required rate of return of 8%; however, Project A's IRR is 7.04%, which is below the required rate of return of 8%. Consequently, only Project B meets the IRR decision rule.

   B is incorrect because the NPV decision rule is to invest if the NPV is greater than 0. Project B's NPV is $1.12 million, while Project A's NPV is –$0.38. Consequently, only Project B meets the NPV decision rule.

3. A is correct. The forecasted timing of cash flow is an important factor affecting both projects' IRR and NPV. In the case of delaying all future cash flows from both projects by six months, the result would be a lower IRR and a lower NPV. Given that all future cash flows are to be received six months later, their present value decreases; hence, the NPV and IRR of an entire project also decrease.

   B is incorrect because when cash flows from the investment projects are delayed by six months, both projects' NPV and IRR would decrease, not stay unchanged.

   C is incorrect because when cash flows from the investment projects are delayed by six months, both project's NPV and IRR would decrease.

4. A is correct. To calculate ROIC, we can use the following formula:

   $$\text{ROIC} = \frac{(1 - \text{Tax rate}) \times \text{Operating profit}_t}{\text{Average invested capital}_{t-1,t}},$$

   where

© CFA Institute. For candidate use only. Not for distribution.

$$\text{Average invested capital}_{t-1,t} = \frac{(\text{Debt} + \text{Equity})_{20X1} + (\text{Debt} + \text{Equity})_{20X2})}{2}$$

$$\text{Average invested capital}_{t-1,t} = \frac{[(25,040 + 39,800 + 6,500) + (25,910 + 40,990 + 6,780)]}{2}$$

$$\text{Average invested capital}_{t-1,t} = 72,510.$$

$$\text{ROIC} = \frac{(1 - 0.22) \times 8,830}{72,510}.$$

ROIC = 9.50%.

5. A is correct. ABC's management aims to achieve a minimum ROIC above 9.20%, but the IRR of both projects fails to meet this investment criterion. Even though Project B has a positive NPV, if it were undertaken, it would reduce ROIC.

   B is incorrect because even though Project B's NPV is $1.12 million (which increases ABC's value to shareholders), its IRR is 8.97%, which is below the target ROIC of 9.20%. Consequently, if Project B was undertaken, it would lower ABC's ROIC.

   C is incorrect because Project A's IRR is below ABC management's target ROIC of 9.2%. Additionally, Project A's NPV is negative, so it decreases ABC's shareholder value.

© CFA Institute. For candidate use only. Not for distribution.

PRACTICE PROBLEMS

The following information relates to questions 1-4

A financial analyst is evaluating the capital structure for Boulder, Inc. Boulder, Inc., is a US-based unleveraged firm with a constant (perpetual) cash flow of $6 million per year before taxes. The firm has a market value of $45 million and a corporate tax rate of 30%. Boulder plans to issue $15 million in debt to retire an equivalent amount of equity. The debt will have a cost of 5.5%.

The financial analyst notes that the tax shield advantage of debt is offset due to the risk of financial distress. He estimates the present value of the cost of financial distress at various debt levels as follows:

Value of Debt	Present Value of Cost of Distress
$5 million	$0.1 million
$10 million	$0.2 million
$15 million	$0.5 million
$20 million	$1.5 million
$25 million	$6.0 million
$30 million	$12.0 million

1. The WACC of the unleveraged firm, prior to the debt issue, is closest to:

   A. 4.00%.

   B. 9.33%.

   C. 13.33%.

2. The use of the $15 million of debt financing increases the value of Boulder by _____ over its unlevered value. Ignore the costs of financial distress.

   A. $4.5 million

   B. $10.5 million

   C. $15 million

3. Including the present value of the cost of distress, the value of Boulder is closest to:

   A. $49 million.

   B. $55 million.

   C. $59.5 million.

4. The level of debt that will maximize the value of Boulder Inc. is closest to:

   A. $15 million.

© CFA Institute. For candidate use only. Not for distribution.

B. $20 million.

C. $30 million.

---

5. A financial analyst is evaluating the capital structure for Plover, Inc., a European-based unleveraged firm with a constant (perpetual) cash flow of EUR10.0 million per year before taxes. The firm has a market value of EUR100.0 million and a corporate tax rate of 20%. Plover plans to issue EUR35.0 million in debt to retire an equivalent amount of equity, so the size of the firm will remain unchanged. The debt will have a cost of 4.5%. Assume the cost of financial distress is close to zero.

   After the debt issuance and change in the capital structure, Plover's WACC is closest to:

   A. 5.47%.

   B. 7.48%.

   C. 8.82%.

© CFA Institute. For candidate use only. Not for distribution.

SOLUTIONS

1. B is correct. The after-tax cash flows for the company are \$6 million × (1 – 0.30) = \$4.2 million. Since the cash flows are assumed to be perpetual, WACC is calculated as follows:

   $r_{WACC} = CF(1 - t)/V$ = \$4.2/\$45 = 9.33%.

   Since this is an all-equity firm, the cost of equity is equal to WACC.

2. A is correct. According to MM Proposition I with corporate taxes, the value of the levered company is greater than that of the unlevered company by an amount equal to the tax rate multiplied by the value of the debt (*tD*), defined as the present value of the debt tax shield.

   $V_L = V_U + tD$ = \$45 million + 0.30(\$15million) = \$49.5 million.

   Thus, the use of debt financing will increase the value of Boulder by \$4.5 million over the unlevered firm. Note that the firm's value increases continuously as additional debt is added to the capital structure, and the optimal capital structure is 100% debt.

3. A is correct. With financial distress included, as debt is added to the capital structure, the levered value of the firm is given by:

   $V_L = V_U + tD - PV$(Costs of financial distress).

   The levered value of Boulder financed with \$15 million in debt is

   V_L = \$45 million + 0.30(\$15 million) – \$0.5 million = \$49 million.

4. B is correct. With financial distress included, as debt is added to the capital structure, the levered value of the firm is given by:

   $V_L = V_U + tD - PV$(Costs of financial distress).

   The following shows calculations of the levered value of Boulder at various debt levels, in \$millions.

D	*PV*(Costs of financial distress)	$V_L = V_U + tD - PV$(Costs of financial distress).
5	0.1	$V_L = 45 + 0.3(5) - 0.1$ $V_L = 46.4$
10	0.2	$V_L = 45 + 0.3(10) - 0.2$ $V_L = 47.8$
15	0.5	$V_L = 45 + 0.3(15) - 0.5$ $V_L = 49$
20	1.5	$V_L = 45 + 0.3(20) - 1.5$ $V_L = 49.5$
25	6.0	$V_L = 45 + 0.3(25) - 6.0$ $V_L = 46.5$
30	12.0	$V_L = 45 + 0.3(30) - 12.0$ $V_L = 42$

   Thus, the optimal amount of debt is \$20 million, which maximizes the value of Boulder at a level of \$49.5 million. Beyond \$20 million in debt, greater leverage

© CFA Institute. For candidate use only. Not for distribution.

reduces the firm value because the present value of financial distress costs more than the offsetting tax benefit.

5. B is correct. The calculations to determine the WACC are as follows:
The after-tax cash flows for the company are EUR10.0 million (1 – 0.20) = EUR8.0 million. Since the cash flows are assumed to be perpetual, WACC for the unlevered firm is calculated as

$r_{WACC} = CF(1 - t)/V$ = EUR8.0 million/EUR100.0 million = 8.0%.

The market value of Plover with the debt financing is EUR107 million:

$V_L = V_U + tD$ = EUR100.0 million + 0.20(EUR35.0 million) = EUR107.0 million.

To find the cost of equity, we first need to find the market value of equity:

$V_L = D + E$, where D and E are market values of debt and equity, respectively.

E = EUR107.0 million – EUR35.0 million = EUR72.0 million.

The cost of equity is given by

$$r_e = r_0 + (r_0 - r_d)(1 - t)\frac{D}{E}$$

= 8.0% + (8.0% – 4.5%)(0.80)(EUR35.0 million/EUR72.0 million) = 9.36%.

Plover's WACC is calculated as follows:

WACC = (Weighting of debt × Cost of debt) + (Weighting of equity × Cost of equity)

= (EUR35 million/EUR107.0 million)(4.5%)(1 – 0.2) + (EUR72.0 million/EUR107.0 million)(9.36%)

= 7.48%, which is lower than the 8.0% for the unlevered firm.

© CFA Institute. For candidate use only. Not for distribution.

PRACTICE PROBLEMS

1. The sequence of processes involved in the creation of a product, both within and external to a firm, including all the steps involved in producing a physical product and delivering it to the end customer, regardless of whether those steps are performed by a single firm, is referred to as the:

   A. value chain.

   B. supply chain.

   C. business model.

The following information relates to questions 2-5

An analyst is reviewing investor presentations from the management teams of two companies with unconventional business models: HealthyPet and DiaSera.

HealthyPet is a pet medical insurer that has a differentiated approach in a business dominated by incumbent property & casualty insurers, for which pet insurance is a niche business relative to business, home, and auto insurance lines. HealthyPet targets the high-income, young, urban, busy professional pet owner segment. It bundles coverage for owners of multiple pets, including exotic pets that incumbents are typically reluctant to insure. The industry incumbents sell insurance policies that reimburse pet owners for a percentage of veterinarian and other medical costs incurred, subject to restrictions and limits. HealthyPet, however, pays veterinarians directly for pet medical costs. It has a preferred network of veterinarians it recommends and for which it negotiates lower prices, but policyholders face relatively few restrictions. It sells through veterinarian referrals and boutique pet stores to maintain an image of exclusivity and prestige and maintains an online storefront for customer convenience. It avoids mass market retail to protect its image. It charges more than its competitors for monthly insurance premiums because it is a better product. The pricing has worked so far because HealthyPet has a strong brand and a distinctiveness that its customers value.

DiaSera designs, makes, and sells a medical device that delivers insulin to people with insulin-dependent diabetes known as an insulin pump. DiaSera's pump is differentiated from those of its competitors in its design and operation; it is far smaller and more aesthetically pleasing than competitors' pumps. It is a single, small device adhered to a user's skin, while competitors' pumps consist of separate handheld controllers attached to a delivery device on a user's skin with tubing.

Insulin pumps are used primarily by people with Type 1 diabetes, which is a smaller market that has complex medical needs, including close management of insulin. The insulin pump market is complex, with several competitors; complement products, such as insulin monitoring devices; and an inexpensive substitute: using multiple daily injections of insulin.

DiaSera is looking to expand its customer reach to include Type 2 diabetics who are on multiple daily insulin injections, which is a market several times the size of the Type 1 diabetics market. DiaSera's pump product is priced at a small premium to competitors. Management says that given significant economies of scale in manufacturing, unit costs will fall substantially as the customer base grows and,

© CFA Institute. For candidate use only. Not for distribution.

even with unchanged pricing, margins will rise significantly. Besides the differentiated product design, the company also offers substantial customer service from highly trained nurses both before and after the sale. Access to customer service is available 24/7 and does not have any additional cost to the customer.

DiaSera makes direct sales to customers and markets to both customers and physicians, because customers must receive a prescription for the pump from a physician before purchase. Products are shipped directly to customers' homes.

2. Complete the grid of business model features for HealthyPet.

Customers
Products
Channels
Pricing strategy

3. Complete the table of business model features for DiaSera.

Customers
Products
Channels
Pricing strategy

4. Discuss key risks, opportunities, and questions regarding the HealthyPet business model.

5. Discuss key risks, opportunities, and questions regarding DiaSera's business model.

© CFA Institute. For candidate use only. Not for distribution.

SOLUTIONS

1. B is correct. A company's supply chain is the network or system inside and outside the company involved in producing a product or service and delivering it to an end user.

   A is incorrect because *value chain* refers to the systems and processes within a firm that create value for its customers. It is the "how" aspect of a business model.

   C is incorrect because the business model is a description of how a business works. It includes descriptions of the core customer base, the product or service that the business offers, the key resources and assets the business uses, and the main partners and suppliers of the business.

2.

Customers	• High-income, young, urban, busy professional pet owners • Own multiple pets and exotic pets
Products	• Insurance policy for pets that are sick or injured • Provides peace of mind to customers • Handles billing directly with veterinarians
Channels	• Veterinarian referrals • Boutique pet stores, online storefront
Pricing strategy	• Premium pricing because it is a premium product • Monthly premiums, similar to subscription

3.

Customers	• Diabetics who need insulin • Expanding customer base to include Type 2 diabetics who are on multiple daily insulin injections
Products	• Differentiated insulin pump for diabetics • Substantial customer service over the phone from highly trained experts both before and after sales • Complex market with complements and substitutes
Channels	• Direct sales to customers • Maintain close relationship with customers • Product shipped directly to customers
Pricing strategy	• Bundling • Premium pricing because it is a premium product and with service

4. There are several apparent risks to HealthyPet's business model. It is not clear that their offering is sufficiently differentiated to build a large-enough customer base to provide the diversification that an insurance company depends on (insurance requires a large pool of policyholders such that there is enough premiums to cover losses). It may incur an adverse selection problem in covering exotic pets because they may be uneconomical to insure as evidenced by the reluctance of established companies to insure them. Finally, if its model is successful, it will likely face competition in the form of imitators, possibly from the incumbent insurers because it is probably possible for them to copy the model of pay veterinarians directly.

   HealthyPet could see opportunity in building a loyal base of customers and a loyal base of veterinarians. Its image of prestige, premium service, and not needing

© CFA Institute. For candidate use only. Not for distribution.

to worry when a pet gets sick could appeal to many pet owners.

Questions include how difficult it would be for other insurers to copy the business model, how HealthyPet plans to establish relationships with veterinarians and how many it has and needs, and whether relationships with veterinarians can be exclusive. It is also questionable whether referrals and boutique pet stores will deliver a sufficient flow of new customers to support the business or whether HealthyPet will instead need to invest in large advertising campaigns directly to consumers.

5. A major risk is the possible incompatibility between DiaSera's pricing strategy and its emphasis on customer service. It is priced only slightly higher than parity with competitors but aims to provide substantially better customer service by using highly trained experts. If customer growth and sales disappoint or take longer than expected, the economies of scale in manufacturing will as well, potentially imperiling profitability or forcing management to raise prices. A second major risk is targeting Type 2 diabetics who have historically not used insulin pumps. While DiaSera's insulin pump seems more convenient than multiple daily injections of insulin, it is more expensive and requires customers to navigate a potentially complicated prescription, purchasing, and service scenario. DiaSera management should discuss key factors that give it confidence in the strategy—such as a segment of customers in the Type 2 market who have tried a pump before, patients who are younger and more likely to try a new product—and establish relationships with key physicians who can recommend the product to patients.

   The clear major opportunities are market share gains in Type 1 diabetes from inferior competitor products and the new market opportunity of Type 2 diabetics who require multiple daily injections. If competitors have not segmented the market to explicitly target this cohort, DiaSera could realize a competitive advantage.

   Key questions include details on its manufacturing strategy, such as estimating unit costs or gross margins at different levels of volume, how experienced the manufacturing management and staff are in this setting, to what extent the company owns the manufacturing resources, and whether there are key suppliers it relies on. Additionally, an analyst will have to develop an outlook for market penetration and sales based on customer segmenting, historical data, precedent product launches, and realistic expectations for customer demand at the current price point. Finally, additional details on how many service professionals are needed per number of customers and at what cost will be essential to modeling operating costs.